

Lecture 10: Pricing strategy

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Lecture overview

- Pricing in Practice
- Alternative Pricing Strategies
- Price Discrimination
- Multiple Product Pricing
- Pricing and the Product Life Cycle

Pricing in Practice

Pricing in Practice

- In textbook models, we often see a single equilibrium price. In the real world, pricing is far more complex and varied.
- The same service can cost vastly different amounts depending on the buyer.
- A single journey to London can have dozens of price points based on:
 - **Timing:** Advance vs. Anytime, Peak vs. Off-peak.
 - **Demographics:** Age (Senior/Child) or Student status.
 - **Add-ons:** Seat reservations or railcard discounts.

Price Discrimination

- Selling the same product to different groups of consumers at different prices.
- To be true price discrimination, the price difference must not be explained by the differences in the cost of supply.

Factors Influencing Real-World Pricing

Why do firms move away from simple $MC = MR$ calculations?

- Market Structure:
 - Monopolies and collusive oligopolies have significantly more price-making power than firms in highly competitive markets.
- Information Gaps:
 - In reality, firms often have scant information regarding their Price Elasticity of Demand.
 - Costs are often estimated rather than precisely known across all output levels, making marginal calculations difficult.

Factors Influencing Real-World Pricing



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- The Aims of the Firm:
 - Not all firms prioritize profit maximization.
 - Pricing changes if the goal is sales maximization, market growth, or long-term brand building.
- The Product Life Cycle:
 - Launch Phase: Prices may be high (skimming) or low (penetration).
 - Maturity/Decline: Prices shift as the product becomes established or is eventually phased out.

Alternative Pricing Strategies

Problems with Marginalist Pricing

In theory, firms equate $MC = MR$ to maximize profit. In practice, this is often impossible due to real-world complexities.

- Lack of Data: Few firms possess the detailed knowledge of costs and revenues required to construct precise curves.
- Uncertainty in Costs: Calculating Marginal Cost (MC) requires knowing exactly how costs shift with every unit of output, difficult in a changing environment (tech shifts, fluctuating input prices).
- Uncertainty in Revenue: Calculating Marginal Revenue (MR) requires knowing the Price Elasticity of Demand.
 - Firms rarely know how many customers they will lose if they raise prices.
 - Under Oligopoly, predicting rival reactions makes identifying a demand curve virtually impossible.
- Managers often rely on guesstimates rather than precise marginalist principles.

The Alternative: Cost-Based Pricing

- Mark-up Pricing (Average-Cost Pricing): Firms calculate the total average cost and add a fixed percentage for profit.

- The Formula:

$$P = AFC + AVC + \text{Profit Mark-up}$$

- The size of the mark-up depends on the firm's specific goals (e.g., maximum profit vs. a stable target).

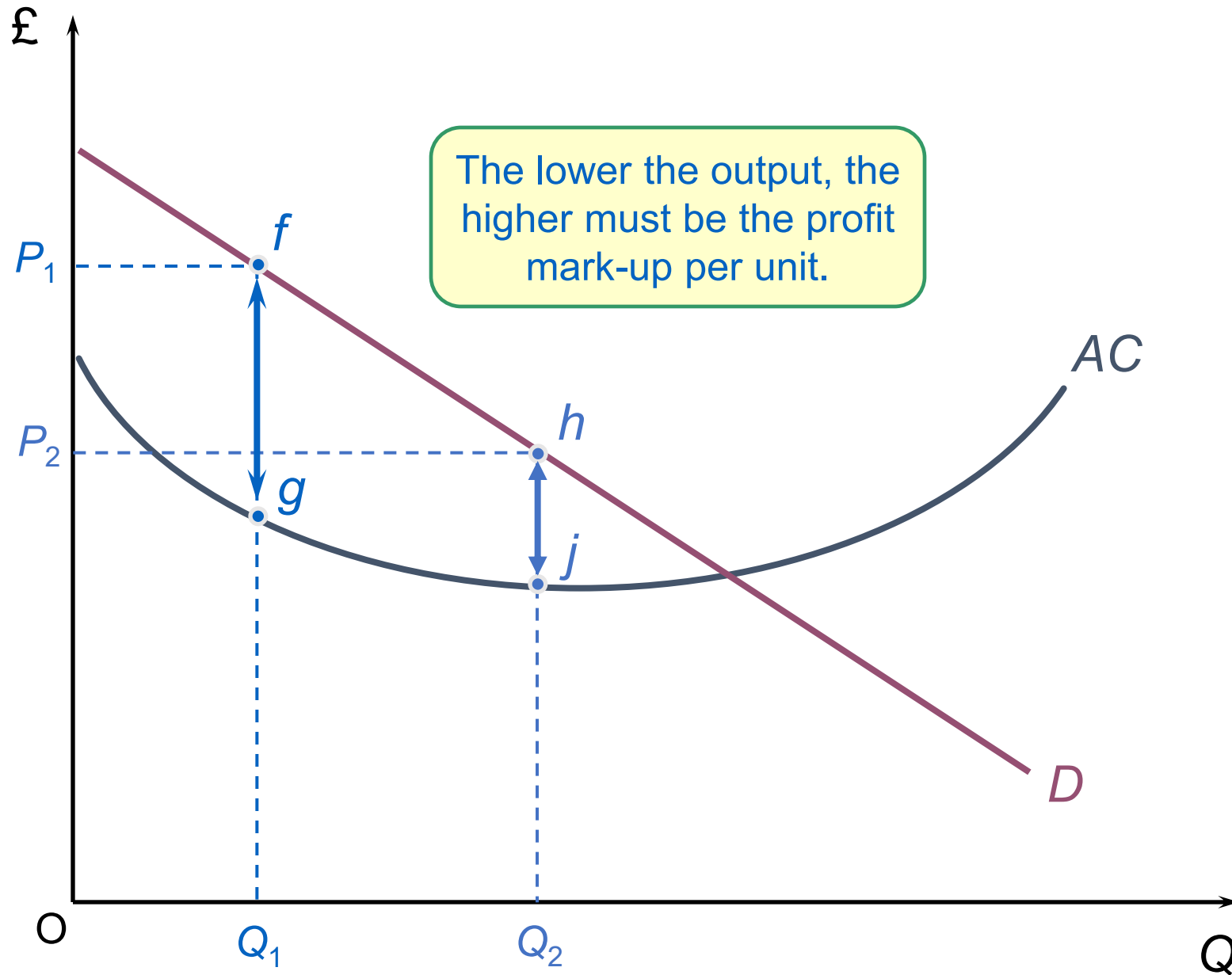
The Alternative: Cost-Based Pricing

- Selecting the right mark-up is a strategic decision influenced by the market environment.

Strategic Considerations

- Market Demand: This is the most significant factor. Even without a precise curve, firms know that higher mark-ups generally lead to lower quantity demanded.
- The Trade-off:
 - High mark-up (fg) = Lower output (Q1).
 - Low mark-up (hj) = Higher output (Q2).

Choosing the output and profit mark-up



The Alternative: Cost-Based Pricing

- Trial and Error
 - The Guessing Game: Prices are often set before sales figures are known.
 - Adjustments: Firms typically use last year's sales as a baseline, add a growth estimate, and then adjust the mark-up based on whether they hit their targets.
- Factors Causing Mark-up Variation
 - Market Growth: Mark-ups are usually higher in expanding markets.
 - Market Power: Monopolies and strong Oligopolies can command higher mark-ups.
 - Economic Downturns: In contracting markets or during periods of rising costs with stagnant demand, firms are often forced to squeeze their mark-up to stay competitive.

Price Discrimination

Price Discrimination

The Logic Behind Price Discrimination

- **The Problem with Uniform Pricing:**
 - Some customers value the product highly and would pay more (the firm misses out on potential revenue).
 - Other customers value it less and won't buy at a high price (the firm misses out on potential sales).
- **The Trade-off:**
 - Raising prices increases profit per unit but reduces the total number of customers.
 - Lowering prices increases sales volume but reduces profit per unit.
- **The Solution:** Charge different prices to different groups based on their **willingness to pay** (valuation/utility).

Price Discrimination

Economic theory defines this practice based on the relationship between price and cost:

- Scenario A (Same Costs): Selling the same or similar product to different customers at different prices when the cost of supply is identical.
- Scenario B (Varying Costs): Selling at different prices where the price gap cannot be fully explained by differences in production or delivery costs.
 - Note: If the price difference perfectly matches the cost difference (e.g., extra shipping fees), it is not price discrimination.

The Three Categories

Economists classify price discrimination into three degrees based on how effectively the firm can target individual consumers:

- **First-degree:** Charging every individual customer the maximum price they are willing to pay (Perfect Price Discrimination).
- **Second-degree:** Charging different prices based on the quantity consumed (e.g., bulk discounts).
- **Third-degree:** Charging different prices to different segments or groups (e.g., student discounts vs. adult prices).

Q Price discrimination, if possible, allows a firm with market power to increase its profits by charging different prices that:

- A. reflect the different marginal costs of producing each unit.
- B. reflect the different average costs of producing each unit.
- ☒ C. reflect the different willingness to pay of each consumer.
- D. cause the demand curve to shift to the right.
- E. cause the supply curve to shift downwards.

First-Degree (Perfect) Price Discrimination



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- First-degree price discrimination represents the theoretical extreme where a firm captures all possible revenue from every customer.
- The seller charges each consumer a unique price equal to their maximum willingness to pay (their reservation price).
- In this model, 100% of the consumer surplus is converted into producer profit.

The Information Problem

- **Asymmetric Information:** Only the buyer knows their true maximum valuation.
- **The Honesty Barrier:** Customers will not truthfully reveal their maximum price if they know it will result in a higher charge.
- Because of these information gaps, perfect discrimination remains more of a theoretical benchmark than a common business reality.

Personalised Pricing in Practice

While perfect discrimination is rare, firms use person-specific pricing to approximate it:

- **Bargaining and Haggling:** Sectors where prices are negotiated allow sellers to size up the customer.
- **Skilled Salesmanship:** Experienced sellers use observation and negotiation to extract as much surplus as possible.
- **Examples:**
 - Second-hand car dealerships.
 - Professional services (lawyers or consultants) using bespoke quotes.

The Cost-Benefit Trade-off

- Negotiating individual prices is incredibly time-consuming.
- A firm must decide if the extra revenue gained from haggling exceeds the extra administrative and labor costs of the process.

Third-Degree Price Discrimination

Third-degree price discrimination involves segmenting the market into distinct groups and charging each group a different price.

- **Group-Based Pricing:** Unlike first-degree discrimination, there is no haggling. The firm sets a fixed price for specific categories (e.g., student discounts, senior rates, or geographic pricing).
- **Direct Price Discrimination:** Along with first-degree, this is considered direct because the firm dictates the price based on consumer attributes rather than letting consumers choose their own pricing tier.

Criteria for Choosing a Grouping Characteristic

To split the market effectively, the chosen trait must meet four essential standards:

- **Observability:** The characteristic must be easy for the firm to verify (e.g., asking for a photo ID or checking a postcode).
- **Informative of Elasticity:** The trait must correlate with Price Sensitivity. Groups with lower price elasticity (less sensitive) are charged higher prices.
- **Legality:** The grouping must comply with local laws.
- **Customer Acceptance:** The basis for the price difference must be perceived as fair to avoid brand damage or boycotts.

Examples of third-degree price discrimination

Characteristic	Example
Age	16–25 or senior rail card; half-price children's tickets in the cinema.
Gender	'Ladies' night' in a bar or club where men pay the full price for drinks while women can get the same drinks at a discounted price (illegal in many countries).
Location	Pharmaceutical companies often charge different prices for the same medicine/drug in different countries. Consumers in the USA are often charged more than those from other countries.
Occupation	Apple, Microsoft and Orange provide price discounts to employees of educational institutions.
Business or individual	Publishers of academic journals charge much lower subscription rates to individuals than university libraries.
Past buying behaviour	Firms often charge new customers a lower price than existing customers for the same product or service as an 'introductory offer'.

The Role of Price Elasticity

- **The Rule:** The group with the lowest price elasticity of demand (least responsive to price changes) is charged the highest price.
- **Uniformity within Groups:** While prices differ *between* groups, every individual *within* a specific group pays the same price.

Why is it More Feasible than First-Degree?

- **Lower Informational Demands:** Firms don't need to know the maximum price for every individual; they only need to understand the average willingness to pay for a broad group.
- **Prevention of Arbitrage:** For this to work, it must be impossible or very difficult for a consumer to switch groups (e.g., an adult cannot easily reclassify themselves as a child to get a cheaper movie ticket).

Q Which one of the following conditions is NOT necessary for (third-degree) price discrimination to take place?

- A.** The firm must be a monopoly.
- B.** The firm must not be a price taker.
- C.** The markets must be separate.
- D.** People buying in the low-price market must not be able to sell to those buying in the high-price market.

Second-Degree (Indirect) Price Discrimination

Unlike first and third-degree models, second-degree price discrimination allows consumers to self-select their pricing tier based on their own preferences and price sensitivity.

- Indirect Pricing: The firm offers a range of options; customers voluntarily choose the one that suits them.
- The Incentive: Used when firms cannot easily observe characteristics like age or location (third-degree) but want to separate price-sensitive customers from price-insensitive ones.

Second-Degree (Indirect) Price Discrimination

Quantity-Based Pricing:

- Quantity Discounts: A lower price applies to all units once a certain volume (Q^*) is reached.
- Block Pricing (Declining Tariff): A lower price applies only to the extra units purchased beyond the initial threshold.

Self-Selection Mechanisms

How do firms get rational consumers to pay more? By making lower prices conditional on effort or timing.

- **Coupons and Vouchers:**

- Targeted at customers with high search costs.
- Only price-sensitive consumers find it worthwhile to spend time collecting and organizing vouchers.

- **Inter-temporal Pricing (Timing):**

- **High-Tech Launches:** Early adopters with low price sensitivity pay a premium for new releases (e.g., the latest iPhone). Prices drop later to capture more sensitive buyers.
- **Airline Seats:** Prices often rise as the flight date approaches. Business travelers (low sensitivity) pay more for last-minute flexibility, while vacationers (high sensitivity) book early for lower rates.

Product Versioning

Firms create different versions of a core product to separate the market based on perceived quality or status.

- **Versioning:** Offering different tiers (e.g., First Class vs. Economy, or Value vs. Finest ranges).
- **The Damaged Goods Strategy:**
 - Firms deliberately remove features or reduce performance of a standard product to create a budget version.
 - It often costs *more* to produce the lower-quality version because of the extra step of damaging or limiting the software/hardware.
- **The Condition for Success:** To keep high-value customers from trading down, the consumer surplus they get from the premium version must be higher than what they would get from the cheap version.

Specialized Pricing Strategies

Beyond the three degrees of discrimination, firms use specific structures to manage demand and maximize revenue.

1. Peak-Load Pricing

- Charging higher prices during periods of peak demand and lower prices during off-peak times.
- **Examples:** Rush hour train fares.
- **Economic Rationale:**
 - **Price Elasticity:** Consumers have lower elasticity at peak times (e.g., commuters must get to work regardless of price). This is genuine price discrimination.
 - **Capacity Constraints:** Higher prices also reflect higher marginal costs as infrastructure reaches its limit.

Specialized Pricing Strategies

2. Two-Part Tariffs

- A pricing system where customers pay two separate charges:
 - **Access Price:** A fixed entry or standing charge.
 - **Usage Price:** A variable price based on the amount consumed.
- **Common Sectors:** Telecommunications (mobile plans) and Energy (gas/electricity).
- **The Firm's Aim:**
 - To secure a guaranteed lump-sum payment per customer, which helps cover high fixed infrastructure costs.
 - To capture more revenue than a simple per-unit price would allow.

Specialized Pricing Strategies

3. Challenges in Competitive Markets

- **The Competition Problem:** Setting the correct balance between the fixed charge and usage rate is difficult when rivals are aggressive.
- **Switching Behavior:** If the fixed fee is too high, customers may easily switch to another supplier.
- **Market Segmentation:** To combat this, firms offer diverse plans:
 - **Standard Two-Part Tariffs:** For medium to high-usage users.
 - **Pay-As-You-Go:** No fixed fee, aimed at light users who prioritize low entry costs.

Conditions Necessary for Price Discrimination

While price discrimination is highly profitable, it is not available to every firm. Three specific conditions must be met for the strategy to be viable.

1. Market Power

- **Price Setting Ability:** The firm must not be a price taker. It must face a downward-sloping demand curve, giving it the discretion to set prices.
- **Competitive Exclusion:** Price discrimination is impossible under Perfect Competition, as any firm attempting to raise prices for one group would immediately lose those customers to rivals.

Conditions Necessary for Price Discrimination

2. Prevention of Resale (Arbitrage)

The strategy fails if consumers in the low-price market can sell the product to those in the high-price market.

- If resale is possible, consumers may buy in bulk at a discount or use a student ID to buy tickets and flip them to full-price customers.
- Platforms like eBay and other online marketplaces have made resale easier, potentially forcing firms to abandon high-price tiers.

Conditions Necessary for Price Discrimination

- **Natural Barriers to Resale:**
 - **Services:** Many products cannot be resold by nature (e.g., haircuts, restaurant meals, or dental work).
 - **Transaction Costs:** Resale is unlikely if the good is inexpensive, perishable, or difficult to transport (the hassle outweighs the profit).
- **Firm Intervention:** If the good is expensive and easy to transport (e.g., electronics), the firm may need to intervene legally or through technology (e.g., non-transferable digital tickets) to block arbitrage.

Conditions Necessary for Price Discrimination

3. Differing Price Elasticity of Demand

- **Segmenting Sensitivity:** For price discrimination to be profitable, different consumers must have different levels of sensitivity to price changes.
- **The Pricing Rule:** The firm will always identify and charge the highest price to the group with the least elastic (less sensitive) demand.
- **Profit Opportunity:** If everyone had the exact same elasticity, the firm would gain nothing from splitting the market; a uniform price would be just as effective.

Q If a supermarket has a 'buy 2 get a 3rd one free' offer, this is an example of:

- A. first-degree price discrimination.
- ☒ B. second-degree price discrimination.
- C. third-degree price discrimination.
- D. demand pricing.
- E. mark-up pricing.

Q Some firms provide coupons and vouchers on their websites. Any customer can print off these vouchers and use them to obtain their next purchase from the firm at a discounted price. This is most likely to be an example of:

- A. first-degree price discrimination.
- ☒ B. second-degree price discrimination.
- C. third-degree price discrimination.
- D. uniform pricing.
- E. It is impossible to say without more information.

Multiple Product Pricing

Multi-Product Pricing Strategies

Most modern firms do not sell just one product; they manage a portfolio of goods with interrelated demand and production costs.

The Multi-Product Reality

- **Distinct Markets:** A firm may produce unrelated items (e.g., a vacuum manufacturer that also makes irons).
- **Product Ranges:** A firm may offer different models within the same market (e.g., vacuums with varying suction power and features).
- **Strategic Coordination:** Each product requires its own price, but the firm must consider how the price of one item affects the sales of another.

Multi-Product Pricing Strategies

Interrelated Demand: Loss Leaders

- Products sold at a dramatically reduced price, often below average cost, to attract customers into a store.
- Attract high foot traffic in the hope that customers will purchase other items with full profit mark-ups.
- **Full-Range Pricing:** Assessing the pricing of the *entire* product portfolio together to maximize net profit rather than individual item profit.
- **Success Factors:**
 - **High Price Elasticity:** The bargain must be sensitive enough to draw in many shoppers.
 - **Strategic Placement:** Positioning loss leaders in specific areas of the store to force customers to pass high-margin items.

Multi-Product Pricing Strategies

Complements and Substitutes: Firms must adjust pricing based on how their own products interact with each other:

- **Complementary Products:** Increased sales of Product A drive sales of Product B.
 - **Example:** Selling an Apple iPhone/iPad increases revenue from the App Store and iCloud subscriptions.
 - **Strategy:** Firms may lower the price of the base product to lock users into a high-margin ecosystem.
- **Substitute Products:** Products that compete with each other within the same firm's range.
 - **Example:** Kellogg's produces multiple types of breakfast cereals; a sale on one may disassemble sales of another.
 - **Strategy:** Prices must be set jointly to ensure the firm maximizes total revenue across the whole brand rather than just shifting sales between its own products.

Pricing and the Product Life Cycle

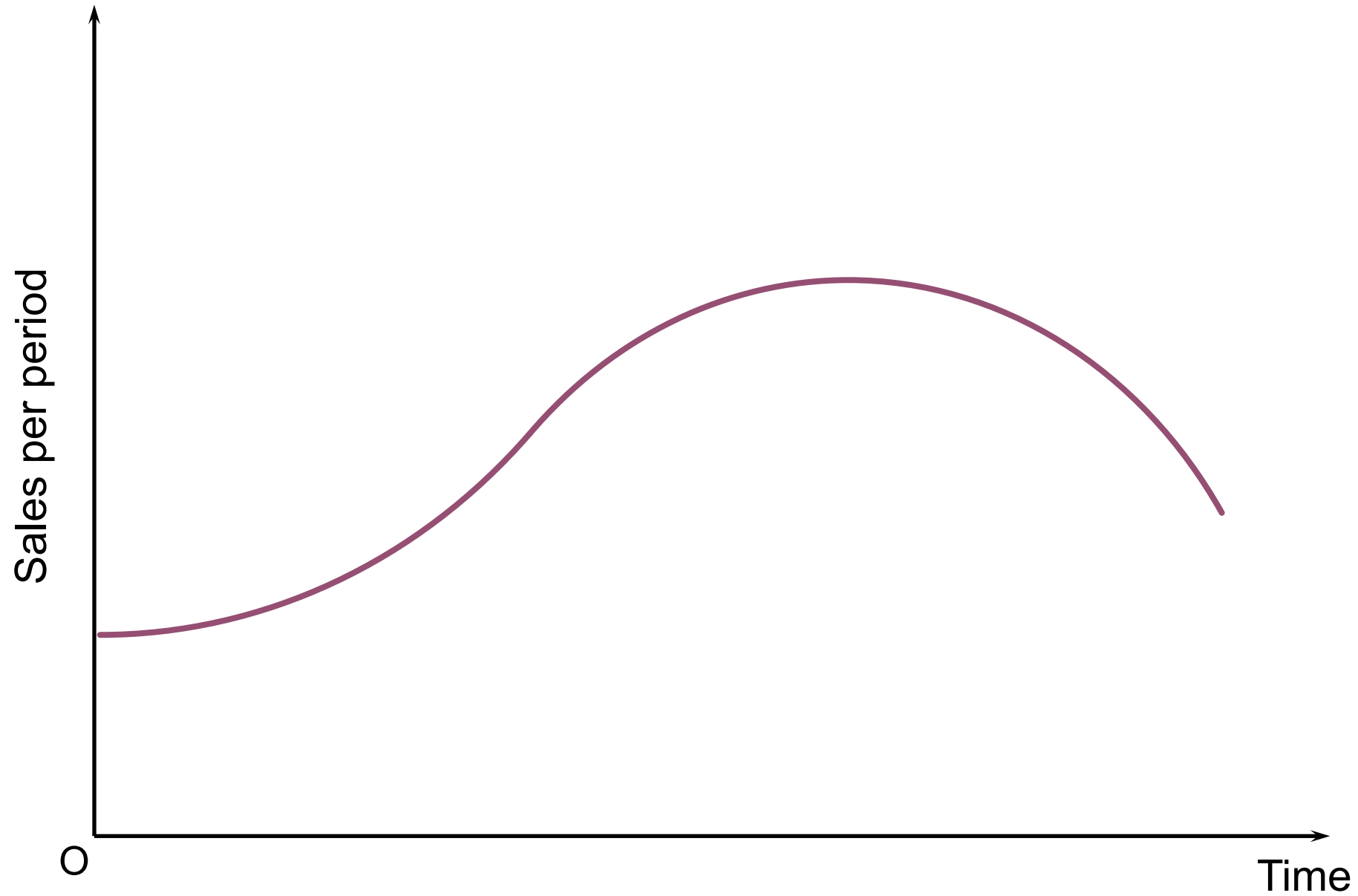
The Product Life Cycle (PLC)

Most products follow a predictable path from launch to obsolescence. Understanding this cycle is critical for setting the right price at the right time.

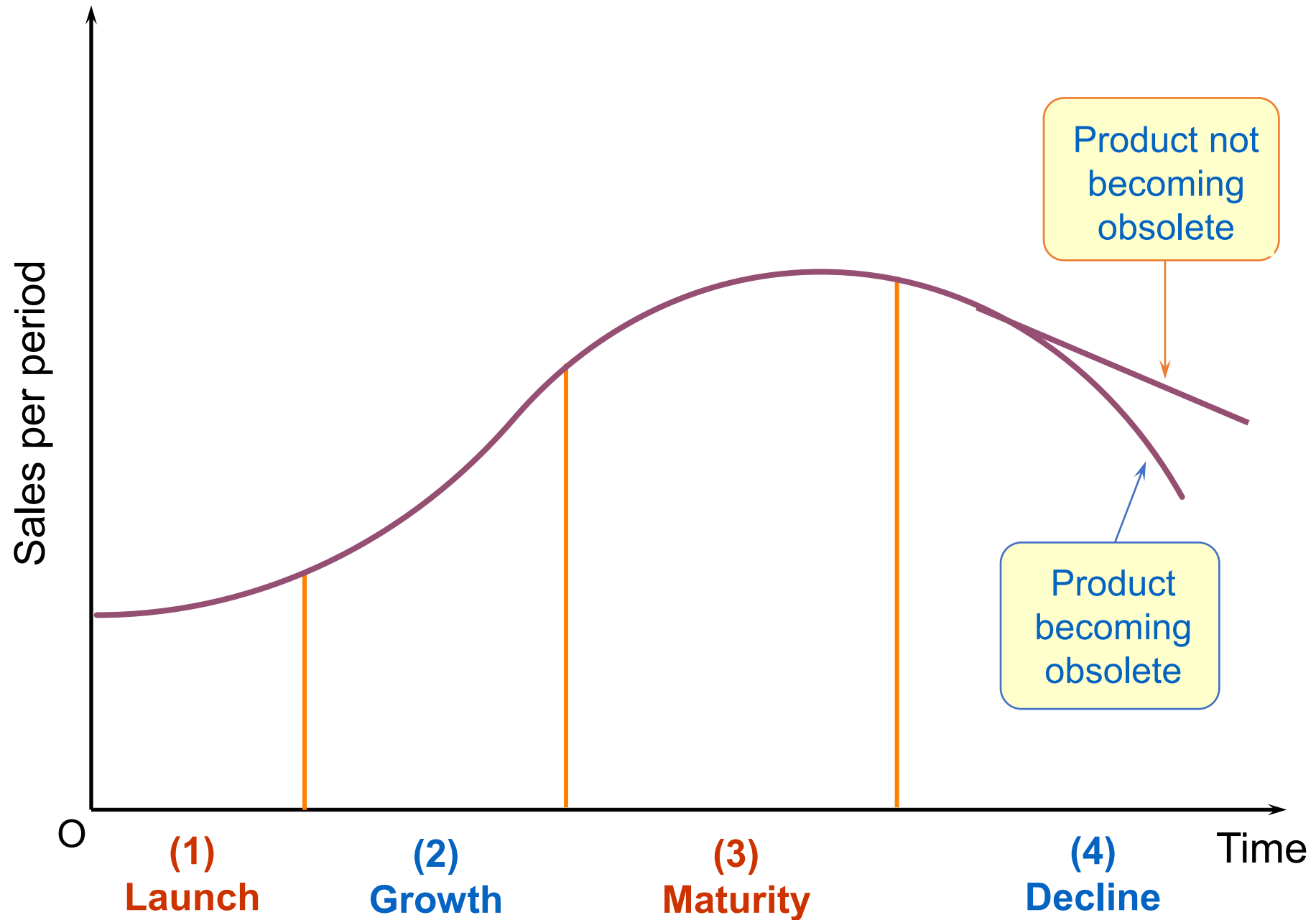
The Four Stages of the PLC

- **Launch:** The product is introduced to the market. (e.g., Electric cars, biodiesel).
- **Growth:** Sales expand rapidly as the product gains mainstream appeal. (e.g., Smartphones, music downloads).
- **Maturity:** Sales level off; the market is well-established. (e.g., Automatic washing machines, DIY products).
- **Decline:** Sales fall due to market saturation or newer technology. (e.g., Audio cassettes, analogue TVs).

The stages in a product's life cycle



The stages in a product's life cycle



Stage 1 – The Launch Stage

Pricing at launch depends on the firm's long-term goals and the uniqueness of the product.

- **Monopoly Power:** If the product is radically new, the firm often has a temporary monopoly with price-inelastic demand.
- **Strategy A: Price Skimming:**
 - Charging high prices initially to maximize profit from early adopters.
 - *Risk:* High profits may attract competitors quickly.
- **Strategy B: Market Penetration:**
 - Keeping prices low to secure a high market share and brand loyalty before rivals arrive.
 - *Goal:* To create a barrier to entry through volume.

Stage 2 – The Growth Stage

- Market Structure: The industry shifts toward Oligopoly as new firms are attracted by rapid sales growth.
- Competitive Dynamics: Because the total market is expanding, all firms can increase sales without aggressive price wars.
- Firms often compete on minor product differences while following the price lead of the original firm.

Stage 3 – The Maturity Stage

- Intense Competition: Sales growth slows down while the number of competitors is at its peak.
- Pricing Strategy: Collusion often breaks down as firms fight for market share.
- Aggressive Pricing: Price wars are common.
- Product Extension: Firms invest in innovation to breathe new life into the product (e.g., adding features to hi-fi systems) to delay the decline stage.

Stage 4 – The Decline Stage

- Market Saturation: Sales fall as consumers only buy replacements or switch to newer technologies.
- The Fight for Survival: Initial competition is fierce; firms use price offers, extended guarantees, and added features to maintain volume.

Two Potential Outcomes:

- Scenario A (Obsolescence): Sales dry up completely and firms exit the market.
- Scenario B (Residual Market): Sales stabilize at a lower level for replacements, returning the industry to a stable, collusive oligopoly.

Q In which stage is it likely that price competition is intense and firms invest in product innovation to stimulate growth in sales?

- A. Launch
- B. Growth
- ☒ C. Maturity
- D. Decline

Questions